Smooth Nonquadratic Affine Maximal Graphs

Radial Integral Certificates, Exact Jets, and Higher-Dimensional Continuation

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Abstract

The October 5, 2026 source constructs a smooth entire affine maximal graph in dimension ten by regularizing a radial system at the origin and continuing it inside a cooperative invariant box. We reconstruct its reduction and prove source-relative refinements: explicit local contraction bounds, a triangular exact-jet recursion with a weighted residual certificate, and a rational invariant box valid in every graph dimension at least ten. The resulting normalized trajectories are strictly increasing and converge to the lower stationary corner; their radial factors are regularly varying with explicitly identified indices. The stationary-exponent quadratic already occurs in the classical singular construction and is not claimed as new. Exact face identities, compact-cylinder Hessian bounds, and finite-time error estimates distinguish algebraic certificates from numerical evidence. An exact global-coordinate formula and the proved asymptotics also establish completeness of the Berwald–Blaschke affine metric for this family, despite the failure of a uniform global Hessian lower bound. Contribution assessment: C4/5, a substantial proved source-relative all-dimensional and affine-completeness extension with quantitative radial certificates, without a historical novelty or blanket source-validity claim.

1 Problem and contribution boundary

For $u \in C^\infty(\mathbb{R}^n)$ with $D^2u > 0$, the classical affine maximal equation is

$$U^{ij}w_{ij} = 0, \qquad U = \det(D^2u)(D^2u)^{-1}, \qquad w = (\det D^2u)^{-(n+1)/(n+2)}. \tag{1}$$

Repeated indices are summed. Positivity throughout this report is pointwise positive definiteness, not a lower bound by one fixed positive matrix on all of $\mathbb{R}^n$.

The source [1] proves existence in dimension ten by taking

$$u(x,t) = \sqrt{F(|x|)^2 + t^2}, \qquad x \in \mathbb{R}^9, \quad t \in \mathbb{R},$$

where $F(0) = 1$, F is smooth radially at zero, and $D_x^2F > 0$. The ansatz and its dimension-independent reduction, the qualitative local integral construction, and the dimension-ten box are inherited. This report adds quantified local certificates, a rational box for every $m = n - 1 \geq 9$, a

monotone-limit proof of regular variation, and completeness of the resulting affine metric. Each addition has a proof below.

The distinction from previous literature matters. Trudinger and Wang's classical paper [2], Section 7, contains the singular dimension-ten profile $\sqrt{|x|^9+t^2}$ and the stationary-exponent quadratic used later here, written in graph dimension n. Their two-dimensional rigidity result and higher-dimensional result under uniform section-convexity assumptions are not contradicted by merely pointwise Hessian positivity. We do not identify that global section condition with an unproved Hessian lower bound.

The October 8, 2026 primary-source check found [4] at arXiv:2608.25330v1, August 26, 2026, with exponent range $\theta \in (0, n/(n+1)]$ for $w = (\det D^2u)^{-\theta}$. Du's related record [3] is arXiv:2210.09127v5, April 10, 2023, and concerns $\theta \in (0,(n-1)/n]$. Neither range includes the classical $(n+1)/(n+2)$. These are metadata and range checks, not independent audits of all their proofs. The present all-dimension extension is relative to the supplied October 5 source, not a priority claim.

2 The source reduction, with all factors retained

Let $m \geq 1$, $n = m+1$, and set

$$k = m+2, \qquad \gamma = \frac{m+3}{m+2} = 1 + \frac{1}{k}.$$

Primes on radial profiles mean d/dr; subscripts s below mean d/ds. For a nonradial F, put $H = D_x^2 F$.

Proposition 2.1 (Inherited reduction)**.** *If $F, P > 0$ are smooth on an open subset of $\mathbb{R}^m$, $H > 0$, and*

$$\det H = FP^{-\gamma}, \qquad \operatorname{tr}(H^{-1}D_x^2 P) + kP/F = 0, \tag{2}$$

then $u(x,t) = \sqrt{F(x)^2+t^2}$ has positive definite Hessian and solves (1) in graph dimension $m+1$.

Proof. Ordinary differentiation, expressed as a quadratic form, gives

$$D^2u = \frac{F}{u}H + \frac{F^2}{u^3}\left(dt - \frac{t}{F}\,dF\right)^2. \tag{3}$$

The first summand is positive in the x directions, and the second is positive on the remaining direction. With $\zeta = t/F$, the pointwise basis $X_i = e_i + \zeta F_i e_t, e_t$ has determinant one and diagonalizes these two blocks. Thus

$$\det D^2u = F^{m+2}u^{-(m+3)}\det H = (F/u)^{m+3}P^{-\gamma}. \tag{4}$$

Since the exponent in (1) is $k/(m+3)$, this gives $w = Ph(\zeta)$, where $h(\zeta) = (1+\zeta^2)^{k/2}$.

It is important to evaluate the ordinary Hessian in the pointwise basis, rather than differentiate the vector fields and forget their variation. Differentiating $t = F\zeta$ twice gives

$$D^2\zeta(X_i,X_j) = -\zeta F_{ij}/F, \quad D^2w(X_i,X_j) = hP_{ij} - (P/F)\zeta h' F_{ij}, \quad w_{tt} = Ph''/F^2.$$

Taking the trace against the inverse diagonal blocks yields

$$\frac{F}{u}\operatorname{tr}((D^2u)^{-1}D^2w) = h\operatorname{tr}(H^{-1}D_x^2P) + (P/F)\big((1+\zeta^2)h'' - m\zeta h'\big).$$

Direct differentiation shows that the last parentheses equal kh, because $k-1-m=1$. The second equation in (2) therefore makes the trace zero. Multiplication by the positive determinant proves (1). □

For radial F, P and $y = F'$, the eigenvalues of H are y' and y/r, the latter with multiplicity $m-1$. Consequently (2), wherever $y, y' > 0$, is exactly

$$y^{m-1}y' = r^{m-1}FP^{-\gamma}, \qquad (y^{m-1}P')' = -kr^{m-1}P^{1-\gamma}, \qquad F' = y. \tag{5}$$

Indeed the second trace is $P''/y' + (m-1)P'/y$, and multiplication by $y'y^{m-1}$ uses the first determinant equation. No factor r^{m-1} or radial multiplicity may be dropped.

3 A quantified integral construction at the origin

Set $s = r^2/2$, $F(r) = f(s)$, $P(r) = p(s)$. For functions on the *two-sided* interval $[-\delta, \delta]$, define the fixed-endpoint dilation average

$$(\mathcal{A}_m g)(s) = \int_0^1 v^{m-1} g(v^2 s)\, dv.$$

The normalized derivative and flux are

$$q = (m\mathcal{A}_m(fp^{-\gamma}))^{1/m}, \qquad j = \mathcal{A}_m(p^{1-\gamma}), \tag{6}$$

and the exact integral map is

$$\mathcal{T}(f,p)(s) = \left(1 + \int_0^s q(a)\, da,\ 1 - k\int_0^s j(a)q(a)^{1-m}\, da\right). \tag{7}$$

All powers in this section are positive real powers. Both components of the state use the maximum product supremum norm.

Theorem 3.1 (Explicit local constants)**.** *Fix $m \geq 1$ and $0 < \beta < 1$, and put*

$$\ell = 1-\beta, \quad u = 1+\beta, \quad q_- = (\ell u^{-\gamma})^{1/m}, \quad q_+ = (u\ell^{-\gamma})^{1/m}, \quad j_+ = \ell^{1-\gamma}/m.$$

Define

$$L_H = \ell^{-\gamma} + \gamma u \ell^{-\gamma-1}, \qquad L_q = \frac{L_H}{m q_-^{m-1}}, \tag{8}$$

$$L_j = \frac{\gamma-1}{m}\ell^{-\gamma}, \qquad L_p = k\left(q_-^{1-m}L_j + (m-1)j_+ q_-^{-m}L_q\right), \tag{9}$$

$$M = \max\{q_+, kj_+q_-^{1-m}\}, \qquad L = \max\{L_q, L_p\}. \tag{10}$$

If $M\delta \leq \beta$ and $L\delta < 1$, then $\mathcal{T}$ has a unique fixed point in the closed ball $\|(f,p) - (1,1)\|_\infty \leq \beta$. It is C^∞ on $(-\delta, \delta)$. Its radial extensions solve (5) *for $0 < r < \sqrt{2\delta}$, with $F, P, y, y' > 0$, $P' < 0$, and*

$$F(0) = P(0) = 1, \qquad D_x^2 F(0) = I, \qquad D_x^2 P(0) = -\frac{k}{m}I.$$

Proof. On the ball, $\ell \leq f, p \leq u$. The average $m\mathcal{A}_m$ has mass one, so $q_- \leq q \leq q_+$; also $u^{1-\gamma}/m \leq j \leq j_+$. The sum of the two absolute partial derivatives of $fp^{-\gamma}$ is at most L_H. The derivative of its positive m-th root is at most $1/(mq_-^{m-1})$, proving the L_q bound. Differentiating $p^{1-\gamma}$, whose average has mass $1/m$, gives L_j. The product kjq^{1-m} then has Lipschitz constant at most L_p. These estimates hold at each argument a, using the input supremum only over $[-|a|, |a|]$.

Integration from zero proves that $\mathcal{T}$ maps the complete closed ball into itself and contracts it by $L\delta$. Its fixed point is first C^1. If f, p are C^h, differentiation under the fixed integrals gives

$$(\mathcal{A}_m g)^{(h)}(s) = \int_0^1 v^{m-1+2h} g^{(h)}(v^2 s)\, dv.$$

On every smaller closed interval the derivatives are bounded, so dominated differentiation applies. The positive root in (6) is smooth. Equation (7) then gains one derivative, proving smoothness by induction.

With $y = rq(r^2/2)$, changing variables $\rho = rv$ in (6) gives

$$y^m = m \int_0^r \rho^{m-1} F(\rho) P(\rho)^{-\gamma}\, d\rho, \qquad y^{m-1} P' = -k \int_0^r \rho^{m-1} P(\rho)^{1-\gamma}\, d\rho.$$

Differentiation proves (5), with $y' > 0$ and $P' < 0$. At zero, $q(0) = 1$, $j(0) = 1/m$, hence $f_s(0) = 1$, $p_s(0) = -k/m$. The two-sided smooth profiles composed with $|x|^2/2$ give precisely the stated Hessians. □

Corollary 3.2 (A rational dimension-ten radius)**.** *For $m = 9$, $\beta = 1/2$, one may take $\delta = 1/256$ and certify $M \le 4$, $L \le 128$. Thus the contraction factor is at most $1/2$, and the fixed point lies within $1/64$ of $(1, 1)$ in the product supremum norm.*

Proof. For $\gamma = 12/11$, integer-power comparisons give

$$2^\gamma < 5/2, \qquad (3/2)^\gamma < 5/3, \qquad 2^{1/11} < 12/11.$$

Therefore

$$q_-^{-9} \le 10/3, \quad q_-^{-8} \le 3, \quad q_+ \le 5/4, \quad j_+ \le 4/33, \quad L_H \le 235/22, \quad L_j \le 5/198.$$

For the second bound, raise to the ninth power and use $(10/3)^8 < 3^9$. The bound on q_+ follows from $q_+^9 \le 15/4 < (5/4)^9$. Thus $L_q \le 235/66$ and

$$L_p \le 11 \left(\frac{5}{66} + \frac{32}{33} \frac{10}{3} \frac{235}{66} \right) = \frac{75695}{594} < 128, \qquad M \le 4.$$

The map's displacement at every input is at most $M\delta \le 1/64$; the same is true at its fixed point. □

This is a conservative analytic radius, not a largest radius of existence. The dimension-ten source asserts a sufficiently small radius but does not specify these constants.

4 Exact jets and residual certificates

4.1 The dilation moments and a triangular recursion

The operator $\mathcal{A}_m$ acts diagonally on monomials:

$$\mathcal{A}_m(s^h) = \frac{s^h}{m + 2h}. \tag{11}$$

This is an exact integration identity, not quadrature or polynomial closure of the nonlinear equation.

Proposition 4.1 (Exact origin jets)**.** *The smooth local solution has uniquely determined rational Taylor coefficients for every integer $m \geq 1$. Write coefficients without factorials as $f(s) = \sum f_h s^h$, $p(s) = \sum p_h s^h$, and similarly for q, j, with equality understood to any finite Taylor order with its remainder. Start with*

$$f_0 = p_0 = q_0 = 1, \quad j_0 = 1/m, \quad f_1 = 1, \quad p_1 = -k/m.$$

For $h \geq 1$, known $f_0, \ldots, f_h, p_0, \ldots, p_h$ determine

$$q_h = \frac{1}{m}\left\{ \frac{m}{m+2h}[s^h](fp^{-\gamma}) - [s^h]\left(\sum_{\ell<h} q_\ell s^\ell\right)^m \right\}, \tag{12}$$

$$j_h = \frac{[s^h](p^{1-\gamma})}{m+2h}, \qquad f_{h+1} = \frac{q_h}{h+1}, \qquad p_{h+1} = -\frac{k}{h+1}[s^h](jq^{1-m}). \tag{13}$$

In particular,

$$\begin{aligned} f(s) &= 1 + s + \frac{2m+3}{2m(m+2)}s^2 + O(s^3), \\ p(s) &= 1 - \frac{m+2}{m}s + \frac{2m^2-3}{2m^2}s^2 + O(s^3). \end{aligned} \tag{14}$$

Proof. Finite-order Taylor expansion and smooth composition at the positive constant terms are legitimate by Theorem 3.1. Equation (11) gives every coefficient of q^m and j. In q^m, the coefficient containing the unknown q_h is mq_h, since $q_0 = 1$; all other terms use lower indices. This gives (12). Equating coefficients in $f_s = q$, $p_s = -kjq^{1-m}$ gives (13). Every power-series coefficient at a specified finite order is a finite algebraic expression in the already known coefficients and rational binomial coefficients. Induction proves rationality and uniqueness, without asserting convergence of the infinite series.

At order one,

$$q_1 = \frac{2m+3}{m(m+2)}, \qquad j_1 = \frac{1}{m(m+2)}.$$

Thus $f_2 = q_1/2$ and

$$2p_2 = -k\,(j_1 + (1-m)q_1/m) = \frac{2m^2-3}{m^2},$$

which proves (14). □

For $m = 9$, this reads

$$F(r) = 1 + \frac{r^2}{2} + \frac{7}{264}r^4 + O(r^6), \qquad P(r) = 1 - \frac{11}{18}r^2 + \frac{53}{216}r^4 + O(r^6).$$

These are jets of the exact smooth solution. Substituting their finite polynomials into a fractional power does not make the entire nonlinear integral equation exact.

4.2 A weighted bound that retains the omitted jet

Theorem 4.2 (Jet-weighted residual enclosure)**.** *Use a ball and constants from Theorem 3.1, and denote its fixed point by $z_* = (f, p)$. Let $\widehat{z}_N$ be the degree-N Taylor polynomial supplied by*

Proposition 4.1, $N \geq 0$, and suppose it lies in the same admissible ball. If a certified number E_N satisfies

$$\|\widehat{z}_N(s) - \mathcal{T}(\widehat{z}_N)(s)\|_{\mathbb{R}^2,\infty} \leq E_N|s|^{N+1} \qquad (|s| \leq \delta),$$

then

$$\|z_*(s) - \widehat{z}_N(s)\|_{\mathbb{R}^2,\infty} \leq \frac{E_N}{1 - L\delta/(N+2)}|s|^{N+1}. \tag{15}$$

For an arbitrary approximate state $\widehat{z}$ in the ball, the unweighted bound is

$$\|z_* - \widehat{z}\|_\infty \leq \frac{\|\widehat{z} - \mathcal{T}(\widehat{z})\|_\infty}{1 - L\delta}.$$

Proof. The unweighted statement follows by adding and subtracting $\mathcal{T}(\widehat{z})$ and using contraction. The exact solution and its polynomial have equal derivatives through order N. Smoothness therefore makes

$$W = \sup_{0<|s|\leq\delta} \frac{\|z_*(s) - \widehat{z}_N(s)\|_\infty}{|s|^{N+1}}$$

finite. At argument a, every input difference to the averages is at $v^2 a$, hence is at most $W|a|^{N+1}$. The pointwise Lipschitz estimate in the proof of Theorem 3.1 consequently bounds the difference of the two integrands in (7) by $LW|a|^{N+1}$. Integrating from 0 to s, with its orientation, gives

$$\|\mathcal{T}(z_*)(s) - \mathcal{T}(\widehat{z}_N)(s)\|_\infty \leq \frac{LW}{N+2}|s|^{N+2}.$$

Divide the residual comparison by $|s|^{N+1}$, take the supremum, and solve $W \leq E_N + L\delta W/(N+2)$. □

The number E_N must enclose the actual nonlinear integrals and any quadrature error. A formal coefficient cancellation alone is not such an enclosure. A C^0 residual bound also does not bound arbitrary differentiated residuals.

Corollary 4.3 (An explicit linear-jet enclosure)**.** *For $m = 9$, $\delta = 1/256$, and $\widehat{z}_1(s) = (1+s, 1-11s/9)$,*

$$\|z_*(s) - \widehat{z}_1(s)\|_\infty \leq \frac{1408}{15}s^2, \qquad \|z_* - \widehat{z}_1\|_\infty \leq \frac{11}{7680}.$$

Proof. The polynomial lies in the ball of radius $1/2$. The integral-map integrand evaluated at the constant state $(1, 1)$ equals the constant vector $(1, -11/9)$, so integrating it gives $\widehat{z}_1$. At input $\widehat{z}_1$, its difference from that constant integrand is at most $128(11/9)|s|$. Thus $E_1 = 704/9$ is valid. Since $L\delta/(N+2) \leq 1/6$, formula (15) gives $1408/15$. Multiplication by δ^2 gives $11/7680$. □

5 A rational invariant box in every higher dimension

Continue the local radial solution maximally while $r, F, P, y > 0$. With $Z = P'$, this is the ordinary smooth system

$$\begin{aligned} &F' = y, \quad P' = Z, \quad y' = G := r^{m-1}FP^{-\gamma}/y^{m-1}, \\ &Z' = -(m-1)(G/y)Z - kr^{m-1}P^{1-\gamma}/y^{m-1}. \end{aligned} \tag{16}$$

In particular $y' > 0$. For $r > 0$, put

$$A = \frac{ry}{F}, \quad B = \frac{ry'}{y}, \quad C = -\frac{rP'}{P}, \quad D = \frac{C}{B}, \qquad \tau = \log r.$$

Dots below mean $d/d\tau$, not d/dr.

Lemma 5.1 (General logarithmic system)**.** *On this continuation, $A, B > 0$, and*

$$\begin{aligned} \dot{A} &= V_1 = A(1 + B - A), \\ \dot{B} &= V_2 = B(m + A + \gamma BD - mB), \\ \dot{D} &= V_3 = (k - D)A - (m - 1)D + BD(1 - D/k). \end{aligned} \tag{17}$$

Near zero, the exact jets give

$$A = r^2 + O(r^4), \quad B = 1 + \frac{2m + 3}{m(m + 2)} r^2 + O(r^4), \quad D = \frac{k}{m} r^2 + O(r^4). \tag{18}$$

Proof. The first radial equation gives $B = r^m F P^{-\gamma}/y^m$. Logarithmic differentiation of this identity and of A gives the first two equations with $BD = C$. From the second radial equation,

$$\dot{C} = C + C^2 - r^2 P''/P = C(1 + C - (m - 1)B) + kAB.$$

Using $\dot{D} = \dot{C}/B - D\dot{B}/B$ and $1 - \gamma = -1/k$ gives the third equation. Finally, $y = rq(s)$, $y' = q(s) + 2sq_s(s)$, and Proposition 4.1 give (18). These are expansions of smooth functions of s, so they may also be differentiated to obtain the corresponding velocity jets. □

Theorem 5.2 (Rational box certificate)**.** *For every integer $m \geq 9$, set*

$$a = \frac{m}{2}, \qquad b = \frac{m - 2}{2}, \qquad d = \frac{m(m - 5)(m + 2)}{(m - 2)(m + 3)}, \qquad \sigma = \frac{m(m - 9)(m - 1)(m + 1)(m + 2)}{(m - 2)(m + 3)^2}. \tag{19}$$

Then $\mathcal{B}_m = [0, a] \times [0, b] \times [0, d]$ is forward invariant for (17)*, and the normalized local solution lies in it for its whole interval of continuation. Every constant is rational for integer m; for $m = 9$ this is exactly the source box $[0, 9/2] \times [0, 7/2] \times [0, 33/7]$.*

Proof. We have $a > 0, b > 1, d > 0, \sigma \geq 0$, and

$$k - d = \frac{6(m - 1)(m + 2)}{(m - 2)(m + 3)} > 0.$$

The nonzero cross derivatives of V on the box are

$$\partial_B V_1 = A, \quad \partial_A V_2 = B, \quad \partial_D V_2 = \gamma B^2, \quad \partial_A V_3 = k - D, \quad \partial_B V_3 = D(1 - D/k).$$

They are nonnegative. More explicitly, direct polynomial substitution gives the following finite identities on the upper faces:

$$\begin{aligned} -V_1(a, B, D) &= a(b - B), \\ -V_2(A, b, D) &= b(a - A) + \gamma b^2(d - D), \\ -V_3(A, B, d) &= \sigma + (k - d)(a - A) + d(1 - d/k)(b - B). \end{aligned} \tag{20}$$

All their coefficients and box factors are nonnegative. On the lower faces, $V_1(0,B,D) = 0$, $V_2(A,0,D) = 0$, and $V_3(A,B,0) = kA \geq 0$. This proves all six inward inequalities, including their weak zero-margin parts. The upper corner need not be stationary: $V(a,b,d) = (0,0,-\sigma)$.

To justify invariance with weak inequalities, let Π be the coordinatewise projection onto the box. The face signs imply $(z - \Pi z) \cdot V(\Pi z) \leq 0$. In a bounded neighborhood take a Lipschitz constant L_V for V. Along a trajectory in that neighborhood,

$$\frac{d}{d\tau} \operatorname{dist}(z, \mathcal{B}_m)^2 = 2(z - \Pi z) \cdot V(z) \leq 2L_V \operatorname{dist}(z, \mathcal{B}_m)^2.$$

The squared distance to a closed box is continuously differentiable, also where the active face changes. Gronwall's inequality and a first-exit argument prove invariance. By (18), the local state is strictly positive, has $B > 1$, and tends to $(0,1,0)$; hence it starts inside the box. □

Theorem 5.3 (Smooth continuation for every $n \geq 10$)**.** *For every integer $m \geq 9$, the normalized radial solution extends smoothly to all of $\mathbb{R}^m$, with $F, P > 0$ and $D_x^2 F > 0$. The function*

$$u_m(x,t) = \sqrt{F(|x|)^2 + t^2}, \qquad (x,t) \in \mathbb{R}^m \times \mathbb{R},$$

is a smooth nonquadratic entire solution of the classical affine maximal equation in dimension $n = m + 1$.

Proof. Suppose the positive continuation ends at $R < \infty$, and fix $0 < r_0 < R$. On it, the box gives $0 \leq A \leq a$, $0 \leq B \leq b$, and $0 \leq C = BD \leq bd$. Since $d\log F/d\log r = A$, $d\log y/d\log r = B$, $d\log P/d\log r = -C$, integration gives

$$\begin{aligned} F(r_0) \leq F(r) &\leq F(r_0)(R/r_0)^a, \\ y(r_0) \leq y(r) &\leq y(r_0)(R/r_0)^b, \\ P(r_0)(r_0/R)^{bd} \leq P(r) &\leq P(r_0). \end{aligned}$$

Also $|Z| = CP/r \leq bdP(r_0)/r_0$. The state therefore stays in a compact subset of the smooth domain $r, F, P, y > 0$ of (16). Its right side is bounded, so the state has a limit at R in that domain. Ordinary local existence and uniqueness extend it past R, a contradiction.

Thus $F, P, y, y' > 0$ for all $r > 0$. The radial Hessian eigenvalues y' and y/r are positive; at zero it is I. The two-sided local construction supplies smoothness through zero. The two equations (2) hold off the origin and extend to it by continuity. Proposition 2.1 proves the full equation. Finally $u_m(0,t) = \sqrt{1+t^2}$, which is not a quadratic polynomial. □

This is not addition of flat variables to a ten-dimensional graph: the equation's exponent and the radial system both change with m. The rational corner identities above give a self-contained certificate; the argument checks both local and global hypotheses in every dimension under discussion.

6 Monotone normalized profiles and regular variation

Theorem 6.1 (Limit corner and regular variation)**.** *For the solution in Theorem 5.3, all three functions A, B, D are strictly increasing in $\log r$, and*

$$(A,B,D) \longrightarrow (\alpha, \alpha - 1, d_*), \qquad d_* = \frac{(m-1)\alpha - 2m}{\gamma(\alpha - 1)},$$

where

$$\alpha = \frac{m^2 - 2m + 9 - (m+3)\sqrt{(m-9)(m-1)}}{16}, \qquad \pi = \frac{2m - (m-1)\alpha}{\gamma} < 0. \tag{21}$$

In particular, for every fixed $\lambda > 0$*,*

$$\frac{F(\lambda r)}{F(r)} \to \lambda^{\alpha}, \qquad \frac{y(\lambda r)}{y(r)} \to \lambda^{\alpha-1}, \qquad \frac{P(\lambda r)}{P(r)} \to \lambda^{\pi} \quad (r \to \infty). \tag{22}$$

This does not assert a positive finite limit of $F(r)/r^{\alpha}$ *or of* $P(r)/r^{\pi}$*.*

Proof. The smooth expansions (18) show that $V(z(\tau)) = \dot{z}(\tau)$ has every component strictly positive for sufficiently negative finite τ. Along the exact trajectory its velocity v solves

$$\dot{v} = DV(z(\tau))v.$$

The off-diagonal entries of this coefficient matrix are nonnegative by Theorem 5.2. On each finite time interval, adding a sufficient scalar multiple of the identity makes all its entries nonnegative. The integral equation for the exponentially rescaled linear system, or its successive approximations, preserves the positive cone. More precisely each positive component has the lower bound $v_i(\tau_0)\exp(\int_{\tau_0}^{\tau}(DV(z))_{ii}\,dt) > 0$. Therefore all components of z strictly increase for all later finite times.

The rational box bounds them, so they have a finite limit z_∞. By continuity $V(z(\tau)) \to V(z_\infty)$; each component is nonnegative. A positive limiting component would make the corresponding coordinate increase without bound, so $V(z_\infty) = 0$. The limit has $A_\infty > 0$, $B_\infty > 1$, $D_\infty > 0$. Write $A_\infty = \alpha$. The first two stationary equations give

$$B_\infty = \alpha - 1, \qquad D_\infty = \frac{(m-1)\alpha - 2m}{\gamma(\alpha - 1)}.$$

In particular $\alpha > 2m/(m-1) > 2$. For an indeterminate $a > 1$, direct substitution into the third component gives the exact identity

$$V_3\left(a, a-1, \frac{(m-1)a - 2m}{\gamma(a-1)}\right) = \frac{(m+1)(m+2)}{(a-1)(m+3)^2}\,Q_m(a), \tag{23}$$

where

$$Q_m(a) = 8a^2 - (m^2 - 2m + 9)a + 2m^2.$$

Its discriminant is $(m+3)^2(m-9)(m-1)$. For $m > 9$, $Q_m(m/2) = -m(m-9)(m-1)/2 < 0$, so $m/2$ lies strictly between the two roots. Since $\alpha \le m/2$, only the lower root is possible. For $m = 9$ the two roots coincide at $9/2$. This proves (21) and $C = BD \to -\pi$.

For a fixed $\lambda > 0$,

$$\log\frac{F(\lambda r)}{F(r)} = \int_{\log r}^{\log r + \log\lambda} A(e^t)\,dt \longrightarrow \alpha\log\lambda.$$

The same argument with B and $-BD$ proves the other two limits. It also covers $0 < \lambda < 1$, with the oriented integral. □

The polynomial Q_m is exactly the classical polynomial in [2, Section 7, equation (7.11)] after $n = m + 1$. The new assertion proved here concerns the normalized *smooth* positive-origin trajectory

and its limit, not the prior existence of this algebraic relation for a singular power profile. For $m = 9$, $(\alpha, \pi) = (9/2, -33/2)$; for $m = 10$, $(\alpha, \pi) = (25/8, -15/2)$. No conclusion about existence or rigidity in $2 \le m \le 8$ follows from the absence of a real stationary corner of this form.

For every $\epsilon > 0$, convergence of the logarithmic derivatives also gives eventual two-sided power bounds with exponents $\alpha \pm \epsilon$, $\alpha - 1 \pm \epsilon$, and $\pi \pm \epsilon$, after normalizing at a sufficiently large radius. It does not give fixed leading coefficients. In fact $A < \alpha$ implies F/r^α is decreasing, but its limit could be zero; likewise P/r^π is increasing and need not have a finite limit. Slowly varying corrections are not discarded.

7 Finite certificates and their geometric limits

7.1 Exact and inexact face identities

For a fixed integer m, (20) is a finite rational certificate: coefficient comparison proves each identity, and the listed coefficients and factors prove its sign everywhere on the entire face. Checking the vector field at finitely many sampled points is not an alternative proof of those inequalities.

The following residual form explains how more complicated candidate regions could be certified without mistaking approximate polynomial identities for exact ones. Parameterize one compact face by coordinates ξ, let $g(\xi)$ be the inward component of the *true* vector field, and suppose

$$g - \mu = \sum_\ell h_\ell z_\ell^{\mathsf{T}} Q_\ell z_\ell + e, \qquad 0 \le h_\ell \le G_\ell, \quad Q_\ell \succeq -\eta_\ell I, \quad \|z_\ell\|_2^2 \le B_\ell, \quad |e| \le \epsilon_e. \tag{24}$$

All bounds must hold on the full face, with any coefficient, quadrature, or model error retained in e.

Proposition 7.1 (Residual-aware face certificate)**.** *Under* (24)*,*

$$g \ge \mu - \epsilon_e - \sum_\ell \eta_\ell B_\ell G_\ell.$$

If this lower bound is nonnegative on each of the six faces of a box, the true vector field makes that box forward invariant.

Proof. Each quadratic form is at least $-\eta_\ell \|z_\ell\|^2$; multiply by its nonnegative factor and sum, retaining $e \ge -\epsilon_e$. The six nonnegative inward bounds imply the projection inequality used in Theorem 5.2, which proves invariance. □

This is a direct specialization of the residual-aware Gram proof in the actual private original [7]. For (20) the displayed factorization already has zero residual and nonnegative coefficients, so no numerical SDP or sum-of-squares search is required. An interval containing zero does not prove a coefficient is zero.

Proposition 7.2 (A finite-time corridor, not exact invariance)**.** *Suppose a locally Lipschitz vector field V has outward component at most $\eta_i \ge 0$ on either face perpendicular to coordinate i of a box $\mathcal{B}$. Let L_V be its Euclidean Lipschitz constant on a convex neighborhood of that box. For an absolutely continuous approximate trajectory $\dot z = V(z) + e(\tau)$, with $\|e(\tau)\|_2 \le \varepsilon$, as long as it stays in that neighborhood,*

$$d(\tau) \le e^{L_V(\tau - \tau_0)} d(\tau_0) + (\|\eta\|_2 + \varepsilon) \frac{e^{L_V(\tau-\tau_0)} - 1}{L_V}, \qquad d = \operatorname{dist}(z, \mathcal{B}).$$

At $L_V = 0$, the quotient means $\tau - \tau_0$.

Proof. Writing $w = z - \Pi z$, the face assumptions imply $w \cdot V(\Pi z) \le \sum_i |w_i|\eta_i \le \|w\|_2 \|\eta\|_2$. The squared-distance derivative, the Lipschitz bound, and Cauchy–Schwarz give $(d^2)'/2 \le L_V d^2 + (\|\eta\|_2 + \varepsilon)d$ almost everywhere. Apply Gronwall to $\sqrt{d^2 + \nu^2}$, retaining the resulting harmless regularization term, and let $\nu \downarrow 0$. Equivalently the upper Dini derivative of d obeys the corresponding linear differential inequality. Integration gives the displayed bound. □

The neighborhood and the time interval are part of this certificate. Here time is $\log r$. A bound on a finite interval of log radius does not automatically prove global continuation. Moreover the exact box is not robust under arbitrary positive error: adding εe_1 to the vector field makes the A-component strictly outward at its upper corner, where the exact V_1 is zero. Thus replacing exact zero-margin face identities by tiny numerical residuals changes the conclusion.

7.2 Compact Hessian bounds and failure of a global one

Proposition 7.3 (A certified local cylinder)**.** *For the $m = 9$ normalized solution, set $R = 1/\sqrt{128}$. On $|x| \le R$, $D_x^2 F \succeq I/20$. For every finite $T \ge 0$, put $U_T = \sqrt{9/4 + T^2}$. On the compact cylinder $|x| \le R,\ |t| \le T$,*

$$D^2 u \succeq \frac{\min\{1/(40U_T),\, 1/(4U_T^3)\}}{(1 + 5RT/2)^2}\, I. \tag{25}$$

There is no positive constant c such that $D^2 u \succeq cI$ on all of $\mathbb{R}^{10}$.

Proof. The local solution is smooth through the endpoint $s = \delta$: the profiles are positive there and the regular radial ODE extends them locally. All closed-interval bounds pass to that endpoint. From Corollary 3.2, $1/2 \le F, P \le 3/2$, $q \le 5/4$, $q^{-9} \le 10/3$, and $q^{-8} \le 3$. The tangential eigenvalue q is greater than $1/20$. The radial equation gives, without differentiating an approximate profile,

$$y' = \frac{FP^{-\gamma}}{q^8} \ge \frac{(1/2)(3/5)}{(5/4)^8} > \frac{1}{20}.$$

At zero the Hessian is I.

In (3), $u \le U_T$, $F/u \ge 1/(2U_T)$, and $F^2/u^3 \ge 1/(4U_T^3)$. Also $\|\nabla F\| = y = rq \le 5R/4$, so $\|(t/F)\nabla F\| \le 5RT/2$. The shear $(\xi, \eta) \mapsto (\xi, \eta - (t/F)\nabla F \cdot \xi)$ has inverse norm at most $1 + 5RT/2$, by the triangle inequality for identity plus its rank-one off-diagonal part. Apply this lower singular-value bound to the two positive summands in (3); it proves (25). Finally

$$u_{tt}(0, t) = (1 + t^2)^{-3/2} \longrightarrow 0.$$

Testing a proposed uniform lower bound on e_t disproves it. □

Completeness is a different question from a uniform Hessian bound. The Berwald–Blaschke affine metric convention is $g = (\det D^2 u)^{-1/(m+3)} D^2 u$, as in [2, equation (1.2)], with $n = m + 1$. The global coordinate $\omega = \operatorname{arsinh}(t/F(x))$ gives the exact identity

$$g = P^{1/k}\big(D_x^2 F + F\, d\omega^2\big). \tag{26}$$

Indeed (4) gives $(\det D^2 u)^{-1/(m+3)} = (u/F)P^{1/k}$. Also $dt - (t/F)dF = F\, d\zeta$ for $\zeta = t/F$, and $F^3 u^{-2} d\zeta^2 = F\, d\omega^2$. Substitution into (3) proves (26). The inverse coordinate map is $t = F(x)\sinh\omega$, so it is a smooth diffeomorphism on all of $\mathbb{R}^{m+1}$, not just a local chart.

Theorem 7.4 (Complete affine metric of the constructed family)**.** *For every $m \geq 9$, the graph in Theorem 5.3 is complete for its Berwald–Blaschke affine metric. More precisely, there is $c_m > 0$ such that, in the global coordinates (x, ω),*

$$g \succeq c_m\big(|dx|^2 + d\omega^2\big).$$

It is also complete for its induced Euclidean graph metric. Neither assertion supplies a uniform positive lower bound for D^2u in the original coordinates (x, t).

Proof. For $r > 0$ define the positive radial coefficients

$$a_0(r) = P(r)^{1/k}\frac{y(r)}{r}, \qquad b_0(r) = P(r)^{1/k}F(r).$$

Since $B > 1$, the radial Hessian eigenvalue $y' = By/r$ exceeds the tangential eigenvalue y/r. Equation (26) therefore implies

$$g \succeq a_0(r)|dx|^2 + b_0(r)d\omega^2.$$

The exact origin jets give $a_0(0) = b_0(0) = 1$ by smooth extension; at $x = 0$ the same bound follows from $D_x^2F(0) = I$. By Theorem 6.1, the logarithmic derivatives satisfy

$$\frac{d\log a_0}{d\log r} = B - 1 - \frac{BD}{k} \longrightarrow \alpha - 2 + \frac{\pi}{k} = \frac{4\alpha - 6}{m+3} > 0,$$
$$\frac{d\log b_0}{d\log r} = A - \frac{BD}{k} \longrightarrow \alpha + \frac{\pi}{k} = \frac{2m + 4\alpha}{m+3} > 0.$$

Here $\alpha > 2$, and $\pi/k = [2m - (m-1)\alpha]/(m+3)$. Each logarithmic derivative is eventually bounded below by half its positive limit. Integrating from a fixed large radius shows that both $a_0(r)$ and $b_0(r)$ tend to infinity. On the remaining compact radial interval they are continuous and strictly positive, so their common infimum c_m is positive. This proves the asserted global metric inequality. No positive finite leading asymptotic coefficient was needed.

For completeness, a g-Cauchy sequence is Cauchy in the Euclidean coordinates (x, ω), hence converges there to a finite point. On a compact coordinate ball around that point, smoothness bounds g above by a finite multiple of the Euclidean metric. Straight segments in this ball then prove convergence in g-distance. Thus its metric space is complete; equivalently its Riemannian geodesics are complete.

For the induced Euclidean graph metric in the original coordinates, $g_E = I + Du \otimes Du \succeq I$. The identical Cauchy argument, using a local upper bound from smoothness of u, proves completeness. Finally $u_{tt}(0, t) = (1 + t^2)^{-3/2}$ tends to zero in every dimension considered, disproving any uniform Hessian lower bound. □

The affine conclusion uses the global coordinate identity and the strictly positive limiting logarithmic derivatives, not merely pointwise convexity or Euclidean completeness. It is a proved source-relative geometric extension for this constructed family; no historical priority or classification theorem is claimed.

8 Private lineage, assessment, and further work

The actual routed jet proofs [5] distinguish finite polynomial exactness, differentiated row remainders, and recovery under a genuine rank assumption. Their relevant transfer here is explicit: the dilation

moments (11) are exact, the coefficient recursion is triangular at every specified finite order, and the omitted nonlinear remainder stays in Theorem 4.2. A finite jet has not become a global solution. Our fixed integration interval also avoids a moving-endpoint omission; every factor v^{2h} is retained when differentiating the dilation averages.

The complete routed stability chapter [6] was read, including the residual-ball theorem, its radii inequalities, finite perturbation estimate, and coordinate-transfer proposition. Its central usable condition is a verified contraction on a complete admissible ball. Here that space is explicitly $C([-\delta,\delta])^2$ with the maximum norm, not a sampled array, and all fractional powers are bounded away from zero. Local root certification is kept separate from the global invariant box. We do not infer a bounded inverse for an infinite-dimensional equation from a finite Jacobian.

The full original [7] was read, including the replayable global gap and inexact Gram/identity proofs. It motivates (20) as an exact finite sign certificate and Proposition 7.1 as its residual-aware alternative. No general polynomial-optimization runtime theorem is asserted. The exact identities are simpler than invoking a generic hierarchy.

The principal gains are the all-$n \geq 10$ rational continuation certificate, identification of the smooth profile's regular-variation indices, completeness of its Berwald–Blaschke metric, an explicit dimension-ten local radius, and a remainder bound attached to actual radial jets. These are source-specific results with proofs, not merely renamed conventional mathematics. Their ingredients, however, are classical contraction, cooperative ODE comparison, Taylor calculus, and polynomial sign checking. The contribution is assessed C4/5 for the substantial proved all-dimensional family and its additional affine-completeness theorem, which the source explicitly does not assert. This score concerns demonstrated source-relative mathematical extension, not historical priority. The conventional ingredients and unverified priority boundary do not support C5. L4 catalogue applicability is a separate judgment, not the basis of this contribution score.

The limitations are equally concrete. The local radius and constants are conservative. No analyticity or convergence of the full formal jet series is proved. No rate of convergence to the stationary corner or positive finite asymptotic coefficient is supplied. The double root at $m = 9$ is a reason to retain possible slowly varying factors. Pointwise Hessian positivity does not imply uniform global convexity; in this family the latter fails explicitly, even though the affine metric is complete by Theorem 7.4. Classification of all radial solutions and rigidity in smaller dimensions remain outside this report. The positive constant c_m in the completeness proof is not given an effective numerical value.

Useful next steps are tighter interval bounds on the integral map, high-order residual enclosures for the exact jets, and a study of linearized and nonlinear convergence near the lower stationary corner. A rate theorem would need to treat the critical $m = 9$ case separately rather than infer hyperbolic decay from numerical plots. Any numerical continuation should publish both its log-radius interval and the residual budget in Proposition 7.2.

A Literal read scope and provenance

The source `build/source/main.tex` was read in full, lines 1–481, as were `references.bib` lines 1–79 and the preprint README lines 1–18. The substantive blocks are reduction (112–208), local regularization (210–297), logarithmic system (299–369), invariant box (371–427), and global completion (429–475). The source PDF was not visually audited. No verified error was found in these bounded source checks; that is not a global certification of the source or all cited literature.

The private original reads cover `sec-03-jets.tex` lines 1–242, 513–938, and 1211–1305, including the full relevant Leibniz, endpoint, differentiated-remainder, recovery, and nonlinear jet-map proofs; `stability.tex` lines 1–295, complete; and `TR-item-047-astra.tex` lines 1–581, complete. The scaled-norm caller in `foundations.tex` lines 269–305 and report metadata were also read. An initial oversized jet-file retrieval was truncated and is not counted as a complete read. Transitive private sources and the rest of the 1885-line jet file were not systematically audited.

The public primary check read the relevant introduction, metric convention, section-convexity hypothesis, and Section 7 of [2], and the current versioned abstracts/metadata of [3, 4]. It was not an audit of those whole papers. Only public mathematical terms, titles, and URLs were queried; private source content remained local.

The catalogue base is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The frozen October 8 upstream comparison identifies the source family without a text change. The three local text files are independently rehashable against their pinned Git blob IDs; byte identity is not a mathematical verdict. Final report/PDF hashes, actual test counts, native-tool availability, export logs, and all-page review receipts are in the owned audit sidecar and freeze manifest.

SHA-256, main source

`72593e7c344863b0eb11abbf06b982c11b33f3b9a94c0b8d8aa66e931068ee4b`

SHA-256, source bibliography

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SHA-256, preprint README

`318e4a37d6d0727d9d8689ebd6f76fb117392ae2a95178d82acfc61e91a695f5`

SHA-256, original jet chapter

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SHA-256, original stability chapter

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SHA-256, original TR-item-047

`a7686c548ba83c2bdb14a237944c17877b4494edaa0235c56e9194210c9a6c56`

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[5] Muralidhara Subbarao, *Foundations and Catalog of Ram-Localized Integral, Integro-Differential and PDE-Equivalent Equations*, TR1-lin-nonlin, Revision 2, September 2026, private original chapter `sec-03-jets.tex`. Selected actual proofs read, as detailed in Appendix A.

[6] Muralidhara Subbarao, *Ram Calculus for Linear and Nonlinear Integral and Partial Differential Equations: Report 1, Localization, Moment Representations, and Certified Finite Inversion*, September 18, 2026, private original `stability.tex` and the specified scaled-norm caller.

[7] Muralidhara Subbarao, *Beyond Convex Optimization through Ram Calculus: Algebraic Geometry, Moment Panels, and Global Optimality Certificates*, private technical report TR-item-047-astra, September 6, 2026. Complete actual original read.